

# 0/1 Knapsack Problem - Experimental Mode

September 17, 2026

## 0.1 Dynamic Programming Average Execution Time

| Capacity | 10      | 20      | 30      | 40      | 50      | 60      | 70      | 80      | 90      | 100     |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 100      | 0.02261 | 0.03140 | 0.05527 | 0.08635 | 0.09939 | 0.11025 | 0.12320 | 0.13480 | 0.14836 | 0.14145 |
| 200      | 0.01924 | 0.04326 | 0.06289 | 0.08132 | 0.10666 | 0.12357 | 0.14746 | 0.16589 | 0.18771 | 0.21132 |
| 300      | 0.03058 | 0.06456 | 0.09597 | 0.12271 | 0.15272 | 0.18237 | 0.21397 | 0.24275 | 0.28228 | 0.30329 |
| 400      | 0.03981 | 0.08357 | 0.12162 | 0.16318 | 0.20501 | 0.24376 | 0.28064 | 0.32097 | 0.36164 | 0.42520 |
| 500      | 0.06457 | 0.12625 | 0.16742 | 0.21387 | 0.25415 | 0.30336 | 0.35187 | 0.40479 | 0.45734 | 0.50655 |
| 600      | 0.06261 | 0.12438 | 0.18473 | 0.24394 | 0.30415 | 0.35628 | 0.40510 | 0.46496 | 0.52057 | 0.57906 |
| 700      | 0.07026 | 0.13872 | 0.21113 | 0.27419 | 0.34307 | 0.40954 | 0.47523 | 0.53902 | 0.60687 | 0.67093 |
| 800      | 0.07866 | 0.15785 | 0.23235 | 0.31007 | 0.38534 | 0.46151 | 0.53593 | 0.61442 | 0.69354 | 0.76687 |
| 900      | 0.09079 | 0.17888 | 0.26915 | 0.35374 | 0.44091 | 0.53412 | 0.60958 | 0.69782 | 0.78204 | 0.86387 |
| 1000     | 0.09737 | 0.19543 | 0.29176 | 0.38547 | 0.48002 | 0.57709 | 0.66970 | 0.76313 | 0.86244 | 0.93124 |

## 0.2 Simple Greedy Average Execution Time

| Capacity | 10      | 20      | 30      | 40      | 50      | 60      | 70      | 80      | 90      | 100     |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 100      | 0.00366 | 0.00311 | 0.00411 | 0.00572 | 0.00569 | 0.00592 | 0.00593 | 0.00700 | 0.00761 | 0.00641 |
| 200      | 0.00200 | 0.00217 | 0.00255 | 0.00285 | 0.00325 | 0.00359 | 0.00382 | 0.00576 | 0.00546 | 0.00594 |
| 300      | 0.00209 | 0.00277 | 0.00290 | 0.00341 | 0.00380 | 0.00458 | 0.00484 | 0.00650 | 0.00634 | 0.00649 |
| 400      | 0.00214 | 0.00256 | 0.00242 | 0.00351 | 0.00394 | 0.00416 | 0.00460 | 0.00564 | 0.00657 | 0.00832 |
| 500      | 0.00266 | 0.00307 | 0.00325 | 0.00393 | 0.00403 | 0.00538 | 0.00638 | 0.00701 | 0.00785 | 0.00867 |
| 600      | 0.00231 | 0.00261 | 0.00345 | 0.00427 | 0.00419 | 0.00526 | 0.00536 | 0.00608 | 0.00707 | 0.00804 |
| 700      | 0.00201 | 0.00249 | 0.00357 | 0.00377 | 0.00330 | 0.00381 | 0.00409 | 0.00426 | 0.00560 | 0.00674 |
| 800      | 0.00198 | 0.00274 | 0.00314 | 0.00385 | 0.00403 | 0.00489 | 0.00542 | 0.00581 | 0.00694 | 0.00836 |
| 900      | 0.00219 | 0.00313 | 0.00408 | 0.00436 | 0.00450 | 0.00486 | 0.00480 | 0.00522 | 0.00536 | 0.00910 |
| 1000     | 0.00203 | 0.00263 | 0.00311 | 0.00305 | 0.00342 | 0.00374 | 0.00408 | 0.00525 | 0.00576 | 0.00600 |

## 0.3 Proportional Greedy Average Execution Time

| Capacity | 10      | 20      | 30      | 40      | 50      | 60      | 70      | 80      | 90      | 100     |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 100      | 0.00393 | 0.00413 | 0.00762 | 0.01208 | 0.01517 | 0.01818 | 0.02104 | 0.02382 | 0.02831 | 0.02758 |
| 200      | 0.00213 | 0.00278 | 0.00411 | 0.00540 | 0.00874 | 0.01037 | 0.01262 | 0.01524 | 0.01895 | 0.02292 |
| 300      | 0.00252 | 0.00355 | 0.00516 | 0.00676 | 0.00873 | 0.01108 | 0.01444 | 0.01775 | 0.02226 | 0.02655 |
| 400      | 0.00244 | 0.00359 | 0.00445 | 0.00611 | 0.00867 | 0.01075 | 0.01418 | 0.01687 | 0.01975 | 0.03101 |
| 500      | 0.00280 | 0.00424 | 0.00570 | 0.00802 | 0.00885 | 0.01255 | 0.01675 | 0.01970 | 0.02403 | 0.02680 |
| 600      | 0.00247 | 0.00380 | 0.00555 | 0.00875 | 0.01065 | 0.01215 | 0.01577 | 0.01916 | 0.02366 | 0.02851 |
| 700      | 0.00230 | 0.00374 | 0.00624 | 0.00898 | 0.01216 | 0.01486 | 0.01819 | 0.02043 | 0.02247 | 0.02631 |
| 800      | 0.00225 | 0.00389 | 0.00573 | 0.00796 | 0.01067 | 0.01383 | 0.01626 | 0.02065 | 0.02564 | 0.02804 |
| 900      | 0.00241 | 0.00415 | 0.00608 | 0.00866 | 0.01197 | 0.01441 | 0.01706 | 0.02029 | 0.02232 | 0.02883 |
| 1000     | 0.00218 | 0.00372 | 0.00591 | 0.00810 | 0.01086 | 0.01450 | 0.01695 | 0.02110 | 0.02673 | 0.02907 |

#### 0.4 Simple Greedy Accuracy Compared with Dynamic Programming

| Capacity | 10      | 20     | 30     | 40    | 50    | 60    | 70    | 80    | 90    | 100   |
|----------|---------|--------|--------|-------|-------|-------|-------|-------|-------|-------|
| 100      | 100.00% | 2.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 200      | 47.00%  | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 300      | 0.00%   | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 400      | 44.00%  | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 500      | 53.00%  | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 600      | 51.00%  | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 700      | 0.00%   | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 800      | 0.00%   | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 900      | 46.00%  | 28.00% | 40.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 1000     | 65.00%  | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |

#### 0.5 Proportional Greedy Accuracy Compared with Dynamic Programming

| Capacity | 10     | 20    | 30    | 40    | 50     | 60     | 70    | 80    | 90    | 100    |
|----------|--------|-------|-------|-------|--------|--------|-------|-------|-------|--------|
| 100      | 0.00%  | 0.00% | 0.00% | 0.00% | 42.00% | 34.00% | 0.00% | 0.00% | 0.00% | 0.00%  |
| 200      | 51.00% | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 300      | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 43.00% |
| 400      | 0.00%  | 0.00% | 0.00% | 0.00% | 47.00% | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 500      | 54.00% | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 600      | 53.00% | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 55.00% |
| 700      | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 800      | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 900      | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |
| 1000     | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  | 0.00%  | 0.00% | 0.00% | 0.00% | 0.00%  |